CLOUD STORAGE AND DATABASES

A. Introduction to Cloud Storage

❖**Definition:** Cloud storage refers to the practice of storing digital data in logical pools across multiple, often virtualized, servers hosted by a third-party provider, rather than directly on local physical storage devices. Data is accessible over the internet.

CLOUD STORAGE AND DATABASES

- **How it Works:**
- Data is sent to the cloud provider's data centers, where it is stored on interconnected servers.
- The provider is responsible for maintaining the infrastructure, ensuring data availability, and handling backups and security.

KEY CHARACTERISTICS

- **Accessibility:** Data can be accessed from anywhere, anytime, on any device with an internet connection.
- **Scalability:** Users can easily increase or decrease storage capacity on demand, paying only for what they use.
- **Durability and Reliability:** Data is typically replicated across multiple servers and locations to ensure high availability and prevent data loss.
- **Cost-Effectiveness:** Reduces the need for upfront capital expenditure on hardware and ongoing maintenance.
- **Collaboration:** Facilitates easy sharing and collaborative work on files.

B. TYPES OF CLOUD STORAGE

Object Storage:

- **Description:** Stores data as objects in flat namespaces (buckets) rather than in a hierarchical file system. Each object includes the data itself, a unique identifier, and metadata.
- **Use Cases:** Ideal for unstructured data like images, videos, backups, archives, web content, and big data analytics.
- **Examples:** Amazon S3, Google Cloud Storage, Azure Blob Storage.
- **Advantages:** Highly scalable, cost-effective for large datasets, resilient, and good for static content.

FILE STORAGE

- **Description:** Stores data in a hierarchical file and folder structure, similar to traditional Network-Attached Storage (NAS). Data is accessed using file-level protocols like NFS (Network File System) or SMB (Server Message Block).
- **Use Cases:** Suitable for applications that require shared file access and traditional file system semantics, such as enterprise applications, content management systems.
- **Examples:** Amazon EFS, Azure Files, Google Cloud Filestore.
- **Advantages:** Familiar interface, easy to integrate with existing applications.

BLOCK STORAGE

- **Description:** Stores data in fixed-size blocks, each with a unique identifier. These blocks are raw storage volumes that can be attached to virtual machines (VMs) as if they were local hard drives.
- **Use Cases:** Best for high-performance applications, databases (e.g., relational databases, NoSQL databases), and operating systems that require low-latency and high I/O operations.
- **Examples:** Amazon EBS, Azure Disk Storage, Google Persistent Disk.
- **Advantages:** High performance, flexible for various operating systems and applications.

C. CLOUD DATABASES

❖ **Definition:**

- Databases provided as a service by cloud providers, handling the underlying infrastructure, scaling, patching, and backups.
- This allows users to focus on application development rather than database administration.

BENEFITS OF CLOUD DATABASES

- **Managed Service:** Reduced operational overhead (no need to manage hardware, software, backups, patching).
- **Scalability:** Easily scale compute and storage resources up or down as needed.
- **High Availability:** Built-in redundancy and failover mechanisms to ensure continuous operation.
- **Cost-Effective:** Pay-as-you-go model, often more efficient than maintaining on-premise databases.
- **Global Distribution:** Easily deploy databases across multiple regions for low latency and disaster recovery.

TYPES OF CLOUD DATABASES

1. Relational Databases (RDBMS as a Service):

- **Description:** Traditional SQL databases managed by the cloud provider. Support structured data, ACID properties (Atomicity, Consistency, Isolation, Durability).
- **Examples:**
  - **Amazon RDS:** Supports MySQL, PostgreSQL, Oracle, SQL Server, MariaDB, Aurora.
  - **Azure SQL Database:** SQL Server compatible.
  - **Google Cloud SQL:** Supports MySQL, PostgreSQL, SQL Server.
- **Use Cases:** Online transaction processing (OLTP), traditional business applications, financial systems.

TYPES OF CLOUD DATABASES

2. NoSQL Databases (Non-Relational Databases):

- **Description:** Databases designed for flexible schemas, massive scalability, and handling large volumes of unstructured or semi-structured data.
- **Categories:**
  - **Document Databases:** Store data in flexible, semi-structured documents (e.g., JSON).
    - *Examples:* Amazon DynamoDB, Azure Cosmos DB (document API), Google Cloud Firestore.
    - *Use Cases:* Mobile apps, content management, user profiles.
  - **Key-Value Stores:** Simple data model with key-value pairs.
    - *Examples:* Amazon DynamoDB (key-value API), Redis, Memcached.
    - *Use Cases:* Caching, session management.

TYPES OF CLOUD DATABASES

2. NoSQL Databases (Non-Relational Databases):

- **Wide-Column Stores:** Store data in tables with rows and dynamic columns.
  - *Examples:* Apache Cassandra (often self-managed on cloud VMs), Azure Cosmos DB (Cassandra API).
  - *Use Cases:* Large-scale data analytics, IoT data.
- **Graph Databases:** Optimized for storing and querying relationships between data points.
  - *Examples:* Amazon Neptune, Azure Cosmos DB (Gremlin API).
  - *Use Cases:* Social networks, recommendation engines, fraud detection.

TYPES OF CLOUD DATABASES

3. Data Warehousing and Analytics Databases:

- **Description:** Optimized for analytical queries over large datasets, often column-oriented for faster aggregation.
- **Examples:** Amazon Redshift, Google BigQuery, Azure Synapse Analytics.
- **Use Cases:** Business intelligence, reporting, data mining.

D. DATA MIGRATION TO CLOUD STORAGE AND DATABASES

3. Strategies:

- **Lift and Shift:** Migrating existing applications and databases to the cloud with minimal changes.
- **Re-platforming:** Modifying applications to take advantage of cloud-native services (e.g., switching from on-premise SQL Server to Amazon RDS).
- **Re-architecting:** Redesigning applications to be fully cloud-native, often using microservices and serverless architectures.

D. DATA MIGRATION TO CLOUD STORAGE AND DATABASES

- **Considerations:**
  - **Data Volume and Velocity:** How much data needs to be moved and at what rate?
  - **Downtime Tolerance:** How much downtime is acceptable during migration?
  - **Data Consistency:** Ensuring data integrity during and after migration.
  - **Security and Compliance:** Protecting data during transit and at rest in the cloud.
  - **Network Bandwidth:** Adequate network capacity for large data transfers.
  - **Migration Tools:** Using native cloud provider tools (e.g., AWS DMS, Azure Database Migration Service) or third-party solutions.

TOPIC II. SECURITY IN THE CLOUD

- **A. Introduction to Cloud Security**
- **Definition:** Cloud security refers to the set of policies, technologies, applications, and controls utilized to protect virtualized IP, data, applications, services, and the associated infrastructure of cloud computing.
- **Shared Responsibility Model:** A fundamental concept in cloud security.
  - **Cloud Provider (e.g., AWS, Azure, Google Cloud):** Responsible for "security *of* the cloud" (e.g., physical security of data centers, network infrastructure, virtualization layer, underlying compute, storage, and database services).
  - **Cloud Customer:** Responsible for "security *in* the cloud" (e.g., data encryption, network configurations, identity and access management, operating system patches, application security). The extent of customer responsibility varies by service model (IaaS, PaaS, SaaS).

TOPIC II. SECURITY IN THE CLOUD

B. Key Cloud Security Concerns

1. Data Loss and Leakage:

- **Causes:** Accidental deletion, malicious attacks, insecure APIs, insider threats, inadequate encryption.
- **Mitigation:** Data encryption (at rest and in transit), robust access controls, data loss prevention (DLP) solutions, regular backups, secure data handling policies.

2. Insecure APIs and Interfaces:

- **Risk:** APIs are the primary means of interacting with cloud services. Vulnerabilities in APIs can lead to unauthorized access, data breaches, and system compromise.

TOPIC II. SECURITY IN THE CLOUD

B. Key Cloud Security Concerns

- **Mitigation:** API security best practices (e.g., strong authentication, authorization, input validation, rate limiting, logging and monitoring), using secure API gateways.

3. Account Hijacking:

- **Risk:** Attackers gaining unauthorized access to cloud accounts through phishing, stolen credentials, or weak authentication.
- **Mitigation:** Multi-Factor Authentication (MFA), strong password policies, regular security audits, monitoring for unusual account activity, Identity and Access Management (IAM) best practices.

TOPIC II. SECURITY IN THE CLOUD

4. Malicious Insiders:

- **Risk:** Employees, contractors, or cloud provider staff with malicious intent or negligence causing harm.
- **Mitigation:** Strict access controls (least privilege), segregation of duties, robust logging and auditing, background checks, data encryption.

5. Shared Technology Vulnerabilities:

- **Risk:** Vulnerabilities in the underlying infrastructure (hypervisors, shared network components) that can impact multiple tenants.
- **Mitigation:** Cloud providers are responsible for securing the underlying infrastructure. Customers should ensure their applications and OS are patched and secured.

6. Compliance and Governance:

- **Risk:** Meeting regulatory requirements (e.g., GDPR, HIPAA, PCI DSS) when data is stored and processed in the cloud.
- **Mitigation:** Understanding shared responsibility, leveraging cloud provider compliance certifications, implementing appropriate controls, maintaining audit trails.

C. CLOUD SECURITY BEST PRACTICES AND TECHNOLOGIES

1. **Identity and Access Management (IAM):**
   a) **Principle of Least Privilege:** Granting users and services only the minimum permissions required to perform their tasks.
   b) **Role-Based Access Control (RBAC):** Assigning permissions based on job functions.
   c) **Multi-Factor Authentication (MFA):** Adding an extra layer of security for user authentication.
   d) **Federated Identity:** Integrating on-premise identity systems with cloud IAM for seamless access.

C. CLOUD SECURITY BEST PRACTICES AND TECHNOLOGIES

2. Data Encryption:

a) **Encryption at Rest:** Encrypting data stored on disks (e.g., using KMS - Key Management Service).

b) **Encryption in Transit:** Encrypting data during transmission over networks (e.g., TLS/SSL).

c) **Key Management:** Securely managing encryption keys (e.g., using Hardware Security Modules - HSMs, or managed key services).

C. CLOUD SECURITY BEST PRACTICES AND TECHNOLOGIES

3. Network Security:

a) **Virtual Private Clouds (VPCs):** Isolating cloud resources in a logically isolated network.

b) **Security Groups/Network ACLs:** Firewall rules to control inbound and outbound traffic to instances.

c) **Firewalls and Web Application Firewalls (WAFs):** Protecting applications from common web exploits.

d) **Intrusion Detection/Prevention Systems (IDS/IPS):** Monitoring network traffic for suspicious activity.

e) **VPNs/Direct Connect:** Securely connecting on-premise

C. CLOUD SECURITY BEST PRACTICES AND TECHNOLOGIES

4. Application Security:

- **Secure Software Development Lifecycle (SSDLC):** Incorporating security throughout the application development process.
- **Vulnerability Scanning and Penetration Testing:** Regularly testing applications for vulnerabilities.
- **Code Review:** Identifying security flaws in application code.

5. Security Monitoring and Logging:

- **Centralized Logging:** Collecting logs from all cloud resources for security analysis.
- **Security Information and Event Management (SIEM):** Aggregating and analyzing security logs for threat detection.
- **Cloud Security Posture Management (CSPM):** Continuously monitoring cloud configurations for misconfigurations and compliance deviations.

C. CLOUD SECURITY BEST PRACTICES AND TECHNOLOGIES

6. Incident Response:

- **Planning:** Developing a comprehensive incident response plan for cloud environments.
- **Automation:** Automating incident detection and response where possible.
- **Forensics:** Capability to collect and analyze forensic data in the cloud.

Q/A

NEXT TOPIC

Monitoring, Logging, and Performance in Cloud

CAT 1 NEXT WEEK (ONLY 1 HOUR)

END OF THE LECTURE

THANK YOU